Milky Way Structure Final Project

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Understanding the structure and dynamics of the Milky Way requires a comprehensive characterisation of its stellar populations in both spatial and kinematic dimensions. This project presents a two-part analysis based on a high-quality cross-matched dataset from Gaia DR3 and APOGEE-2 DR17. In the first module, stars are structurally classified into canonical Galactic components—bulge, thin and thick discs, stellar halo, and the solar neighbourhood—using metallicity, Galactocentric coordinates, and vertical scale height. Validation through Toomre diagrams reveals substantial overlap between components, highlighting the limitations of purely spatial or chemical selection schemes.
The second module extends the analysis into the dynamical domain through the computation of full space velocities. Velocity ellipsoid projections and stellar density maps reveal global disk properties, a significant vertex deviation, and the presence of radial streaming motions. Kinematic substructure is further investigated through chemo-dynamical analysis in the solar vicinity, with partial identification of moving groups using unsupervised clustering. The Galactic rotation curve is derived between 6–12 kpc, and spatially resolved maps of the Oort constants provide local measurements of shear, vorticity, and divergence. Overall, the results support the need for multidimensional selection criteria to robustly distinguish stellar populations and to characterise the non-axisymmetric features of the Galaxy.

1. INTRODUCTION

Understanding the structure and dynamics of the Milky Way is essential for constraining the processes that govern galaxy formation and evolution. As the closest laboratory for Galactic archaeology, the Milky Way enables the study of stellar populations across a wide range of spatial and temporal scales. In recent years, the advent of high-precision astrometric surveys such as *Gaia* and complementary spectroscopic programmes like APOGEE has made it possible to probe both the spatial distribution and full kinematics of stars with unprecedented accuracy.

This study is structured in two complementary modules that exploit the synergy of *Gaia* DR3 and APOGEE-2 DR17 data to investigate the structural and dynamical properties of the Galactic stellar population. The first module focuses on the spatial classification of stars into the canonical components of the Galaxy—bulge, thin and thick discs, stellar halo, and solar neighbourhood—based on metallicity, Galactocentric radius, and vertical height above the plane. A set of selection criteria was applied to assign stars to components, followed by validation via the Toomre diagram to assess the coherence of the resulting subsets.

The second module extends the analysis to the dynamical domain by computing the three-dimensional space velocities (U, V, W) in a Galactocentric reference frame. These velocities were used to explore the kinematic structure of the Galaxy through density maps, velocity ellipsoid projections, and spatial vector fields. Special attention was given to identifying kinematic substructures in the solar neighbourhood, including vertex deviation, radial streaming motions, and moving groups. The rotation curve of the Galaxy was derived in the radial range $6 < R_{\rm GC} < 12$ kpc, and spatially resolved maps of the Oort constants were constructed to probe local shear and vorticity in the velocity field.

Together, these modules provide a multidimensional characterisation of the Milky Way's disc and halo, integrating spatial, chemical, and kinematic information. The results reveal both general agreement with theoretical expectations and local deviations that underscore the complexity of Galactic structure. This integrated approach offers a foundation for more refined studies of Galactic dynamics and population synthesis in future work.

2. METHODOLOGY

A. Data Quality Selection

The APOGEE-2 dataset, part of the Sloan Digital Sky Survey (SDSS), provides high-resolution infrared spectroscopy, offering stellar parameters, radial velocities, and chemical abundances. The data were retrieved from the SDSS database. In contrast, Gaia DR3 delivers high-precision astrometric measurements, including positions, proper motions, and parallaxes, which are critical for distance estimation. The dataset was obtained through a cross-match with APOGEE-2 using TOPCAT.

The data cleaning process aims to achieve three main objectives. First, background noise must be sufficiently reduced, as all astronomical objects exhibit random radiation fluctuations. Once this is accomplished, the required precision and resolution to distinguish individual objects must be determined, ensuring reliable characterisation based on their physical properties and minimising systematic errors. Finally, a threshold for radial velocity scatter must be set to enable accurate kinematic studies.

This section introduces three key quality parameters that ensure reliable distance and velocity measurements.

B. Signal-to-Noise Ratio (S/N)

The signal-to-noise ratio (**S/N**) is a statistical measure of how well an object is measured, typically defined as the ratio between the signal, S (the number of photons detected from the source), and the noise, N (the total random contribution from various sources affecting the measurement). Different types of noise can affect the measurement, depending on their origin. These include background or sky noise, dark noise (due to thermal radiation from the detector itself), read-out noise (from the amplifier), and shot noise, among others. Since these noises are considered independent, a generalized expression for the S/N is obtained by adding all these noise sources in quadrature [1, 2].

In practice, a higher S/N indicates a clearer and more reliable spectrum. A common threshold is set at 100 [3]. To determine whether this is an adequate limit for this project, a statistical review has been conducted. As presented in **Table 1**

Table 1. Statistics for S/N values [a]

Mean	Median	Q_1	Q_3
203.18	140.18	92.81	239.58

[a]Calculated through TOPCAT .

Using , the histogram of the S/N provided by the cross-matched data shows a positively skewed distribution (**Figure 1**), with a mean of 203.18, suggesting that, on average, the observations are of high quality. On the other hand, the median is 140.18, meaning that a threshold close to this value would include only half of the data. Moreover, the Q_1 value of 92.81 indicates that 75% of the data have an S/N higher than this. Therefore, in order to minimize the data loss and maximize the S/N, a threshold of 120 should be sufficient.

This threshold was applied using 's Row Subsets feature, by selecting stars with SNR $\geq$ 120.

C. Renormalised Unit Weight Error (RUWE)

The *Renormalised Unit Weight Error (RUWE)* is a parameter from *Gaia DR3* that indicates the quality of the astrometric fit for a star in the catalogue. It is defined as the ratio of the *Unit Weight Error* (UWE), which measures the error in the astrometric fit of Gaia and the expected value of **UWE** for a star with a given magnitude and colour ($\mathbf{UWE_{expected}}$). The ideal value of **RUWE** for a well-behaved star is approximately 1.0. However, in practice, it is recommended to filter out values greater than 1.4, as they may indicate possible binary systems, poorly modelled stars, or astrometric errors [4, 5].

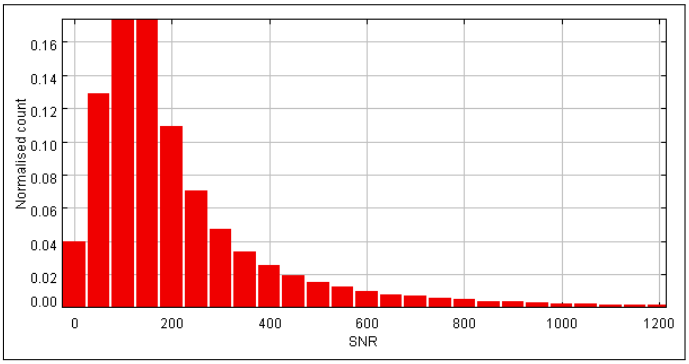


Fig. 1. Histogram of S/N (SNR) values using histogram function.

Sources with RUWE < 1.4 were selected via the Row Subsets function in TOPCAT.

D. Velocity Scatter (VSCATTER)

VSCATTER quantifies the scatter in radial velocity (**RV**) measurements across multiple observations of the same star. A lower VSCATTER value suggests consistent RV measurements, which is crucial for precise kinematic studies. A typical threshold for high-quality data is **VSCATTER** $\leq 0.5\,\mathrm{km\,s^{-1}}$ to exclude stars with large RV uncertainties [6]. Low radial velocity scatter ensures consistency across multiple epochs, critical when tracing orbital motions and Galactic dynamics.

The radial velocity scatter cut (VSCATTER $\leq$ 0.5 km/s) was implemented using a boolean expression within TOPCAT's Row Subsets tool.

After applying the quality cuts, approximately 45% of the original cross-matched sample has been retained, going from a sample size of 682,284 to 308,231 astronomical objects. While this represents a significant reduction in the total number of sources, it ensures a high level of reliability for both astrometric and spectroscopic parameters, which is essential for accurate distance estimation and kinematic analysis.

E. Parallax correction

The primary goal of a second cleaning based on parallaxes is to ensure the reliability of the distance estimates for the stars in the sample. The parallax (ϖ, in arcseconds) is inversely proportional to the distance of the star [7]. For stars that are closer, larger values of parallax are expected, and for stars that are farther away, the parallax values become smaller.

To account for biases or errors introduced by the instrument or measurement process itself a systematic adjustment must be applied to parallax, this is the **zero-point correction**. For Gaia DR3, the zero-point correction is **-0.029 milliarcseconds (mas)**, so that corrected parallaxes are $\varpi - ZPC$ [8]. Using TOPCAT's Synthetic Columns tool, the ZPC value was substracted from the original parallax column values.

As seen in the histogram of corrected parallaxes shown in **Figure 2**, there are negative parallaxes. These appear in the Gaia DR3 catalogue as a natural result of statistical noise and model fitting in the limit of very small true parallaxes or, when parallaxes exceed unphysical values, as a result of cross-matching artefacts. These values are retained in the catalogue to ensure unbiased statistical properties and are not to be interpreted as physically meaningful inverses of distance [9]. Only sources with positive corrected parallaxes were retained.

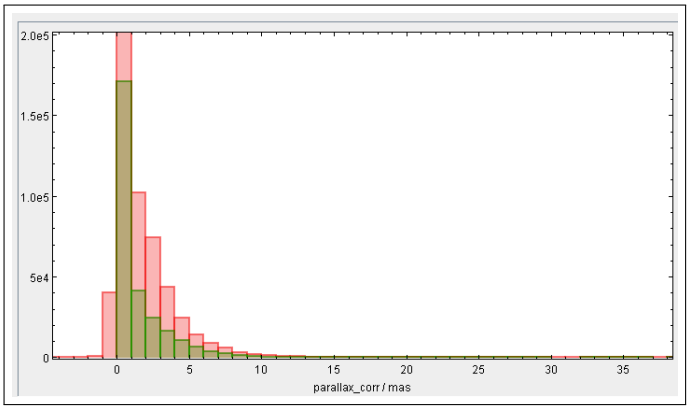


Fig. 2. Histogram of corrected parallaxes for the complete cross-matched data (in red) and the quality selected data (in green).

To assess the reliability of each distance estimate on a per-star basis, the relative parallax errors were defined as the ratio between the original parallaxes errors and the corrected ones $(\sigma_\omega/\omega_{\rm corr})$. Then two quality cuts for the parallax error (10% and 20%) cleaning were compared based on distance, delimiting nearby and distant stars. The Fifth Catalogue of Nearby Stars (CNS5) builds the concept of nearby stars as those within 25 pc of the Sun, using Gaia EDR3 [10]. So distance was calculated in the approximation previously stated as $d_{\rm pc} = \frac{1000}{\omega_{\rm corr}}$.

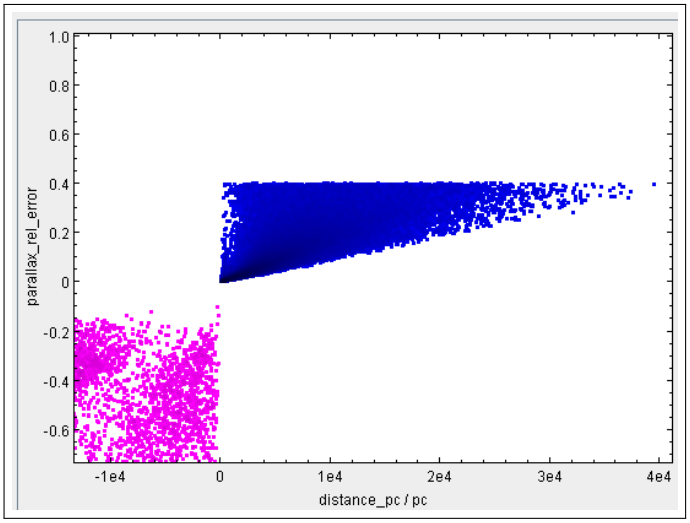


Fig. 3. Corrected parallax error in function of stellar distance for quality selected data without negative parallaxes (blue) and without this restriction (magenta), for both selection schemes.

Then two selection schemes were defined:

- A strict case (10/40): requiring $< 10\%$ parallax error for nearby stars and $< 40\%$ for distant stars.
- A moderate case (20/40): requiring $< 20\%$ for nearby stars and $< 40\%$ for distant stars.

While both 10% and 20% error thresholds were tested for stars within 25 pc, the less strict condition did not increase the number of sources compared to the 10% cut. This indicates that all nearby stars meeting the former criterion already satisfy the more stringent threshold, suggesting a high astrometric precision for the local sample. This can be seen in the **Figure 3**, although it is not explicit because the plots align perfectly and they cannot be distinguished, for both negative and strictly positive selections. The strictly positive selection has a maximum parallax error of approximately 0.5, so both schemes are equivalent.

Therefore, the 10/40 combined selection was adopted as a quality filter, yielding 281,612 sources left.

To summarise, the data cleansing was defined by the following conditions:

$$\left[\begin{array}{ll} \textbf{I.} & \text{SNR} \geq 120 \\ \textbf{II.} & \text{RUWE} < 1.4 \\ \textbf{III.} & \text{VSCATTER} \leq 0.5\ \text{km/s} \\ \textbf{IV.} & \omega_{\rm corr} > 0 \\ \textbf{V.} & \begin{cases} \text{If } d \leq 25\ \text{pc:} & \sigma_\omega/\omega_{\rm corr} < 0.10 \\ \text{If } d > 25\ \text{pc:} & \sigma_\omega/\omega_{\rm corr} < 0.40 \end{cases} \end{array}\right] \quad \textbf{(1)}$$

Although quality cuts reduce systematic uncertainties, they may also introduce selection biases, especially against low-mass or distant stars, which tend to have lower SNR or higher parallax uncertainties.

F. Galactic coordinates

Galactic Cartesian coordinates (X, Y, Z) relative to the Galactic Centre were derived from Galactic coordinates (ℓ, b) and heliocentric distances d using standard trigonometric transformations [11], assuming that the Sun is located at $R_0 = 8.178$ kpc from the Galactic Centre on the X-axis [12]:

$$\begin{aligned} X &= R_0 - d\cos b\cos\ell \\ Y &= -d\cos b\sin\ell \\ Z &= d\sin b \end{aligned} \quad \textbf{(2)}$$

In the cross-matched data, GLAT and GLON correspond to the galactic latitude and longitude, respectively, so three different columns were created $(X_{\rm gal}, Y_{\rm gal}, Z_{\rm gal})$ containing the values of the galactic coordinates.

G. Galactic components overview

The Milky Way galaxy is a barred spiral galaxy (type SBbc), one of the dominant members of the Local Group in the far outer reaches of the Virgo supercluster of galaxies (part of Laniakea supercluster) [13]. It is comprised of the following components.

G.1. Bulge

Classically, the bulge is defined as a round, kinematically hot stellar structure that is distinct from the main disc of a galaxy [14]. Observationally, however, a more flattened, rotationally

Table 2. Typical properties of Galactic structural components.

Component	Age (Gyr)	[Fe/H]	Scale Height	Scale Radius	Structural Properties
Bulge	> 10	> -1.5	∼0.44 kpc	∼1.0 kpc	Boxy/barred; hot with rotational support (pseudo)
Thin Disk	< 5	$\gtrsim -0.5$	∼0.3 kpc	∼2.5 kpc	Thin exponential; cold, fast rotation
Thick Disk	8–12	−1.3 to −0.3	∼1.0 kpc	∼3.5 kpc	Thick exponential; warmer, lagged rotation
Stellar Halo	> 12	< -1.5	Extended (>5 kpc)	—	Spheroidal; pressure-supported, no net rotation
Solar Neighbourhood	Mixed	−0.7 to 0.2	$\lesssim$0.3 kpc	∼8.2 kpc (local)	Local disk sample; rotationally dominated

supported component can also be identified, known as a pseudo-bulge. The key distinction lies in the morphology and kinematics: classical bulges are more spheroidal and dominated by random motions, whereas pseudo-bulges are flatter and primarily supported by rotation [14]. In this study, the bulge is considered to comprise both components.

It is typically modelled by a Sérsic profile with index $n \sim 1-2$, and has a scale length $r_0 \sim 0.5-1$ kpc [13]. The vertical scale height is approximately 0.44 kpc, with a boxy/peanut shape due to bar instabilities. The total mass of the bulge is estimated at $\sim 2 \times 10^{10}\ M_\odot$.

G.2. Solar Neighbourhood

The solar neighbourhood is a representative local volume of the Galactic disk, located at a Galactocentric radius $R_\odot \approx 8.178$ kpc [12]. It contains a mix of thin and thick disk stars, including intermediate-age populations. Stellar kinematics in this region are used to define the Local Standard of Rest (LSR). The solar neighbourhood is typically defined as a cylindrical volume centred on the Sun, with radial extents ranging from 100 to 500 pc and vertical limits from $|Z| \lesssim 0.3$ to 1.5 kpc, depending on the study. For example, Binney & Tremaine (2008) and Casagrande et al. (2011) adopt compact cylinders with $R \leq 200$–300 pc and $|Z| \lesssim 0.3$ kpc for local stellar population studies. Aumer & Binney (2009) extend the vertical range to $|Z| \leq 1.5$ kpc to analyse velocity dispersions [11, 15, 16]. In this study, we adopt the more recent and widely used definition from Bovy (2017), who defines the solar neighbourhood as a cylinder with radius 200 pc and vertical height $|Z| < 0.5$ kpc [17].

G.3. Thin Disk

The thin disk is the dominant component of the Galactic stellar distribution in the solar neighbourhood. It is characterised by a small vertical scale height (∼300-400 pc), an exponential scale length $\sim$ 2.6 kpc, relatively high metallicity ($[Fe/H] \gtrsim -0.5$), and a young stellar population, typically younger than 5 Gyr. Stars in the thin disk exhibit low velocity dispersions and orbit the Galactic Centre with velocities close to the local circular speed, indicating a kinematically cold and rotationally supported population [13].

G.4. Thick Disk

The thick disk is a structurally and kinematically distinct component, characterised by a larger vertical scale height $\sim 1-1.5$kpc ($\sim$ 900 pc near the Sun [13]), a exponential scale length $\sim 2-3.5$ kpc [13], lower metallicity ($[Fe/H] \sim -0.6$, -1.3 to -0.3 [18]), and an older stellar population (8–12 Gyr) [13] than the population of the thin disk. It is generally interpreted as a relic of the early Milky Way, possibly formed through dynamical heating, radial migration, or early mergers. Stars in the thick disk exhibit moderate rotational lag and higher velocity dispersions compared to the thin disk, suggesting a transition between the disk and the halo populations.

G.5. Stellar Halo

The stellar halo is a spheroidal, slightly flattened (axis ratio $\sim$ 0.7) pressure-supported structure composed of very old ($>$ 12 Gyr), metal-poor ([Fe/H] < -2 to -3, peaks at -1.6 [19]) stars [13] and globular clusters (150 known) [20]. It has a stellar mass of $\sim 10^9$ M$_\odot$ [13] and can be modelled by a density law ($\rho \propto r^{-3.5}$) [21]. It extends to large Galactocentric radii ($>$ 50–100 kpc) with a $\sim 2-3$ kpc (core) scale radius [21].

H. Computation of Spatial Velocities

The Galactic space velocity components U, V, W are computed for each star in the cleaned dataset using full six-dimensional phase-space information. The required astrometric parameters include right ascension (RA), declination (DEC), proper motions in right ascension and declination (PMRA, PMDEC), heliocentric distance (derived from corrected parallax), and radial velocity (provided by the `VHELIO-AVG` column in the cross-matched data set). This quantity is already corrected for the motion of the observer with respect to the barycentre and to the Local Standard of Rest (LSR), and is therefore adopted as the best available estimator of the true radial motion of each star.

The computation was implemented in `Python` using the `SkyCoord` class from the `Astropy` package to represent the ICRS-based coordinates and perform the necessary transformations. The stellar positions and velocities were projected into the Galactocentric frame via the `Galactocentric` transformation, using the default parameters from `Astropy`, which assume the Sun's position at $(X, Y, Z) = (8.122, 0, 0.0208)$ kpc and a circular velocity of 232 km/s. Since the radial velocity is already corrected to the LSR, no further subtraction was applied to the V component.

The resulting U, V, W velocities are defined in a right-handed Galactocentric frame, with U positive toward the Galactic centre, V in the direction of Galactic rotation, and W toward the North Galactic Pole. These velocities form the basis for the kinematic analyses presented in the subsequent sections.

3. RESULTS & DISCUSSION

A. Galactic components classification

A comparison between the previously discussed components is shown in **Table 2**. For the classification the metallicity, the scale height and the scale radius were used, creating subsets of stars within the quality selected data in `TOPCAT`. This yielded 86,816 objects within the Thin Disk subset, constituting 30.8% of the high-quality sample, which is consistent with literature expectations of it begin the component with more statistical weight. For

instance, Jurić et al. (2008) and Bovy et al. (2012) estimate that the thin disk contains ∼85–90% of the local stellar population, although the lower percentage found in this study likely reflects the effect of stringent quality cuts and APOGEE's bias toward evolved stars; the Thick Disk comprised 23,370 objects, representing 8.3% of the sample, close to the 5–10% typically reported in density models [22, 23]; the solar neighbourhood makes up 25.9% of the sample with a sample size of 72,923 confirming that the combined Gaia–APOGEE dataset is most representative of the local thin disk; the halo is strongly underrepresented, comprising only 0.22% (633), in line with its expected rarity (less than 1% in the solar cylinder) and the difficulty of detecting faint, metal-poor stars at large distances with precise astrometry and high-resolution spectroscopy; finally, the bulge was found to contribute 0.77% (2,154), much lower than its true mass fraction (∼20% of the stellar mass; Bland-Hawthorn & Gerhard 2016), but this is attributed to both Gaia's reduced astrometric reliability toward the Galactic centre and the limited bulge coverage in APOGEE-2 [24].

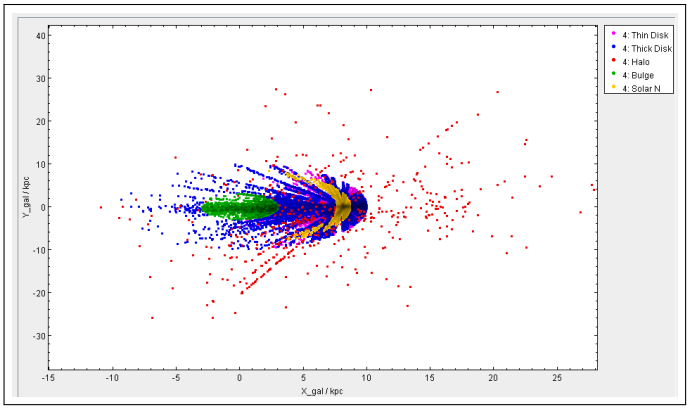


Fig. 4. XY plane positions of the different components. Thin Disk in magenta, Thick Disk in blue, Halo in red, Bulge in Green and Solar Neighbourhood in yellow.

Figure 4 shows the spatial distribution of the classified stellar populations projected onto the Galactic plane ($X_{\rm gal}, Y_{\rm gal}$), in a Galactocentric Cartesian frame. The solar neighbourhood (yellow) forms a compact cluster around the Solar position, as expected from its definition as a local volume sample. On the other hand, the stellar halo (red) exhibits a more random, with no preferred orientation distribution, extending to large radii (> 20 kpc). Its scattered and sparse nature reflects its low density and kinematically hot nature. The bulge population (green) is highly concentrated toward the Galactic Centre, though sparsely sampled due to the limited APOGEE coverage and extinction in that region. To evaluate the two disk components, **Figure 5** presents the distribution of stars in the $R_{\rm GC}$–$|Z|$ plane, where $R_{\rm GC}$ is the Galactocentric radius and $|Z|$ is the absolute vertical distance from the Galactic plane. Here, the thin disk appears tightly concentrated near the plane, with $|Z| < 0.3$ kpc, and spans a radial range from approximately 3 to 10 kpc, as expected for a geometrically thin, rotationally supported population with short vertical excursions. In comparison, the thick disk extends higher above the plane ($0.5 < |Z| < 1.5$ kpc) but occupies a similar radial domain to the thin disk. Moreover, the halo is well-separated from the disks in this diagram. Its stars exhibit high $|Z|$ values, often exceeding 5 kpc, and are spread across a wide radial range; the bulge is compact both vertically and radially, located within $R_{\rm GC} < 3$ kpc and $|Z| < 1.5$ kpc, consistent with its centrally concentrated morphology, and the solar neighbourhood forms a narrow vertical strip centred around $R_\odot$ kpc with very low $|Z|$, as expected from its definition.

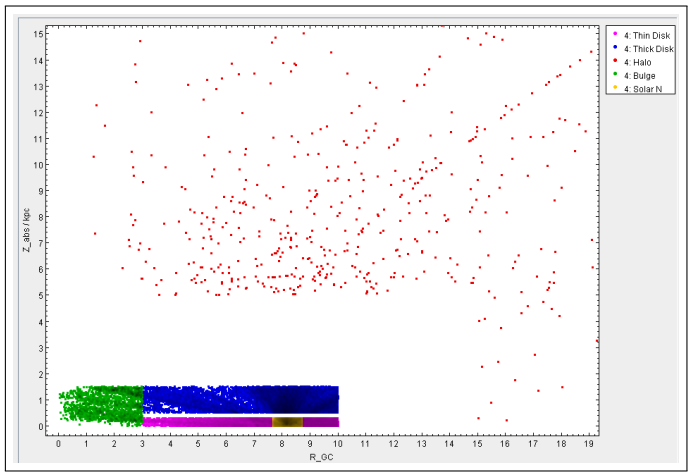


Fig. 5. Galactic centre distance (R_GC) vs height Z.

A.1. Toomre diagram comparison

To put this classification criteria to the test, it was compared against a Toomre diagram, where one can differentiate structures via a velocity criterion. The typical ranges of total space velocities ($V_{\rm tot} = \sqrt{U^2 + V^2 + W^2}$) used to distinguish Galactic stellar populations are well established in the literature. Stars belonging to the thin disk are generally characterised by low kinematic energy, with $V_{\rm tot} < 70$ km/s, reflecting their rotationally supported, dynamically cold orbits. The thick disk is populated by stars with intermediate kinematics, typically within the range $70 < V_{\rm tot} < 180$ km/s, consistent with higher velocity dispersion and a modest lag in azimuthal motion. Finally, stars with $V_{\rm tot} > 180$ km/s are usually assigned to the stellar halo, due to their dynamically hot orbits. These velocity boundaries are consistent with criteria applied in studies using Gaia DR3 kinematics and prior works such as Bensby et al. (2003, 2005) and Nissen & Schuster (2010), providing a robust framework for separating Galactic components purely from their dynamical behaviour [25, 26].

So, in order build the Toomre diagram the heliocentric space velocity components was calculated for each star. The input parameters were right ascension (RA), declination (DEC), distance (previously obtained), proper motions in RA and DEC (from `GAIAEDR3_PMRA` and `GAIAEDR3_PMDEC`), and the heliocentric radial velocity, taken from `VHELIO_AVG` (which, in this case, is already corrected to a baricentric frame and is representative of the Local Standard of Rest). Using the `astropy.coordinates.SkyCoord` class, these quantities were transformed from the ICRS frame to the Galactocentric coordinate system via the `Galactocentric` class. In this fashion, the velocity components (U, V, W) in km/s were obtained.

Since the analysis required velocities relative to the local circular motion, the V component was corrected by subtracting the adopted LSR velocity, here taken as 232 km/s [27]. This yields a corrected azimuthal velocity $V_{\rm LSR} = V - 232\,{\rm km/s}$, allowing to centre the Toomre diagram around the LSR.

Then, the Toomre diagram is produced by plotting $V_{\rm LSR}$ on the horizontal axis and $\sqrt{U^2 + W^2}$ on the vertical axis. This

representation highlights the kinematic temperature of the stars, with lower values of $\sqrt{U^2+W^2}$ corresponding to dynamically cold populations (e.g., the thin disk) and higher values indicating dynamically hot populations (e.g., the halo). There were also overplotted isolines of constant total velocity to guide the interpretation, using the relation:

$$\sqrt{U^2+W^2} = \sqrt{V_{\rm tot}^2 - V_{\rm LSR}^2},$$

where $V_{\rm tot} = \sqrt{U^2+V^2+W^2}$.

The first Toomre diagram (**Figure 6**) shows that, in principle, the classification criteria given in the previous section, while not necessarily robust, gives a good separation for the Solar Neighbourhood, the the thick Disk and the halo, but there is considerable leakage of stars categorize as been part of the Thick Disk into the Halo section; there is a clear concentration of neighbouring stars with low kinetic temperatures inside the 70 km/s region, but a not so predominant presence of the Thick Disk stars in the next region (delimited by the isolines).

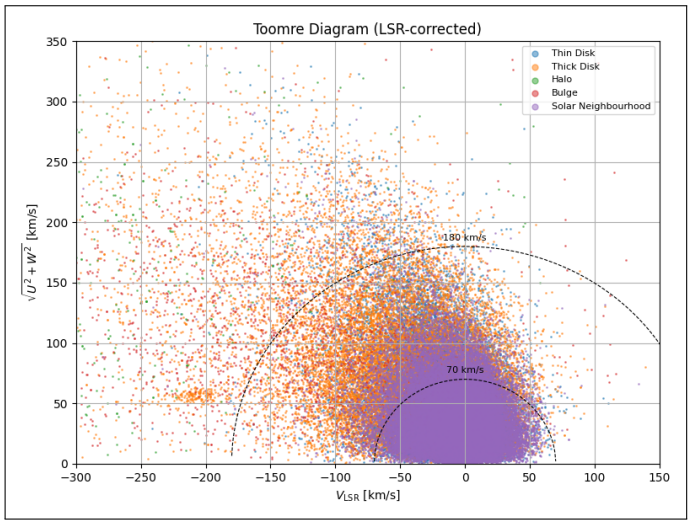


Fig. 6. Toomre diagram, coloured by the classification criteria of **Table 2**. Isolines of total velocity are shown.

The problem is that the inclusion if the Solar Neighbourhood class do not allow to compare directly the two types of Galactic Disks, so **Figure 7** removes this classification.

Upon close inspection, it becomes evident that a non-negligible number of stars classified as part of the thick disk exhibit kinematics more consistent with the thin disk.

This contamination can be attributed to the fact that the classification criteria used until now rely solely on spatial and chemical properties. These parameters are not capturing fully the kinematic differences that also define these structures; metal-poor stars located at moderate vertical heights—which may belong to the thin disk but fall outside its defined parameter ranges, and thus are instead assigned to the thick disk.

To address this issue, future classifications should consider adding a velocity-based criterion instead on relying only in the criteria utilized here. This adjustment would not only reduce the contamination of thick disk stars into the thin disk kinematic regime, but also provide a more robust separation of the disk components.

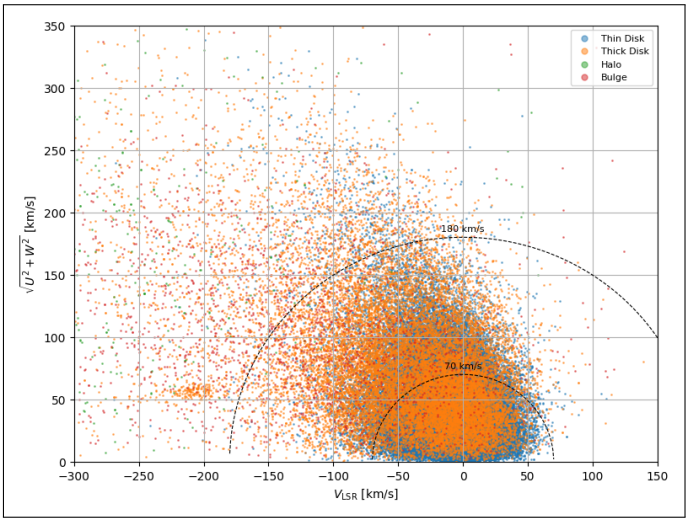


Fig. 7. XY plane positions of the different components. Thin Disk in magenta, Thick Disk in blue, Halo in red, Bulge in Green and Solar Neighbourhood in yellow.

B. Stellar Density Maps

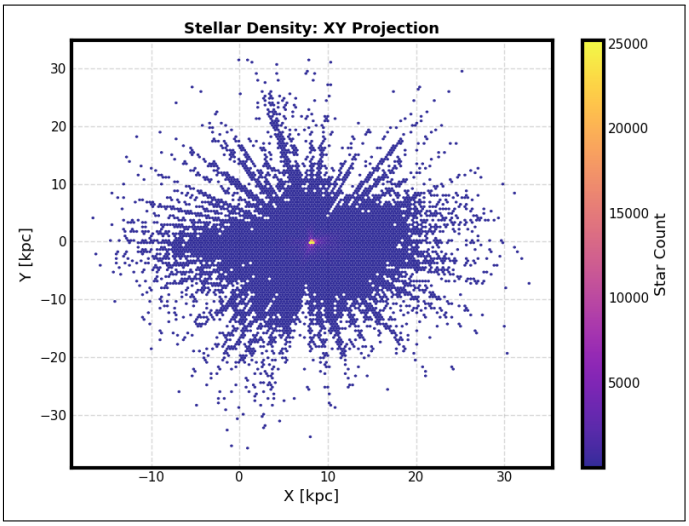


Fig. 8. XY projection of the two-dimensional density map in Galactocentric Cartesian coordinates.

To explore the spatial distribution of stars, two-dimensional density maps in Galactocentric Cartesian coordinates were constructed using hexagonal binning. The three projections—XY, XZ, and YZ—highlight the structural components of the Milky Way and provide insight into survey coverage and geometric selection effects.

In the **XY projection** (Figure 8), the disk structure appears clearly delineated, with a high-density region concentrated around $X \sim 8$–10 kpc and $Y \sim 0$ kpc, consistent with the solar position in the Galaxy. The density smoothly declines radially, but numerous low-density extensions appear, especially in directions aligned with what could be APOGEE's line-of-sight fields. These radial arms suggest structured survey coverage, rather than an isotropic stellar distribution.

In the **XZ projection** (Figure 9), the disk structure is again prominent, with most stars confined within $|Z| < 2$ kpc and

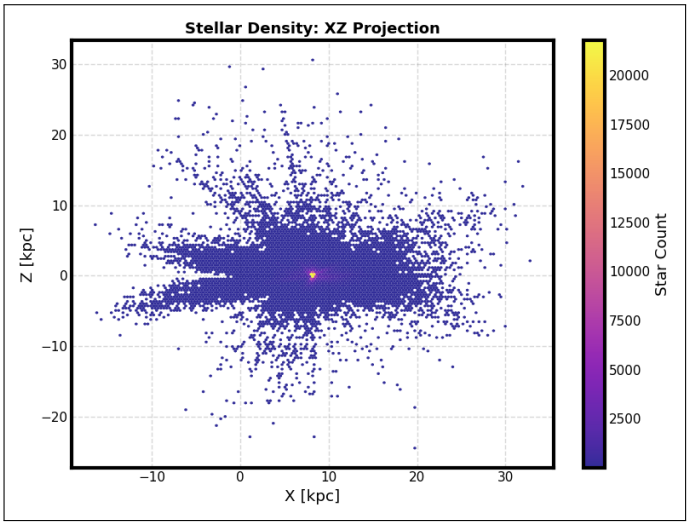


Fig. 9. XZ projection of the two-dimensional density map in Galactocentric Cartesian coordinates.

centred at $Z = 0$. This vertical thickness is consistent with the scale heights of the thin and thick disks, although a small population extends up to $Z \sim \pm 10$–15 kpc, likely corresponding to halo stars. The central overdensity near $X \sim 8$–10 kpc is again visible and corresponds to the solar neighbourhood, heavily sampled in APOGEE.

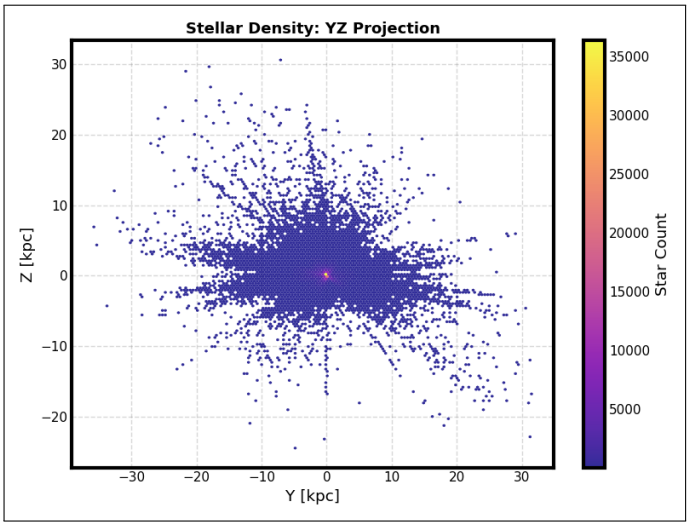


Fig. 10. YZ projection of the two-dimensional density map in Galactocentric Cartesian coordinates.

The **YZ projection** (Figure 10) shows a symmetric disk, flattened in Z, and extending along the Y axis up to $\sim \pm 30$ kpc. The central concentration is prominent, and some higher-altitude structures appear in the outer halo.

C. Velocity Ellipsoid Projections and Vertex Deviation

The global distribution of stellar velocities was analysed using two-dimensional projections of the velocity space in the UV, VW, and UW planes (Figures 11, 12, 13). These projections allow the visualisation of the velocity ellipsoid, revealing the intrinsic dynamical structure of the Galaxy and potential asymmetries associated with non-axisymmetric perturbations or moving groups.

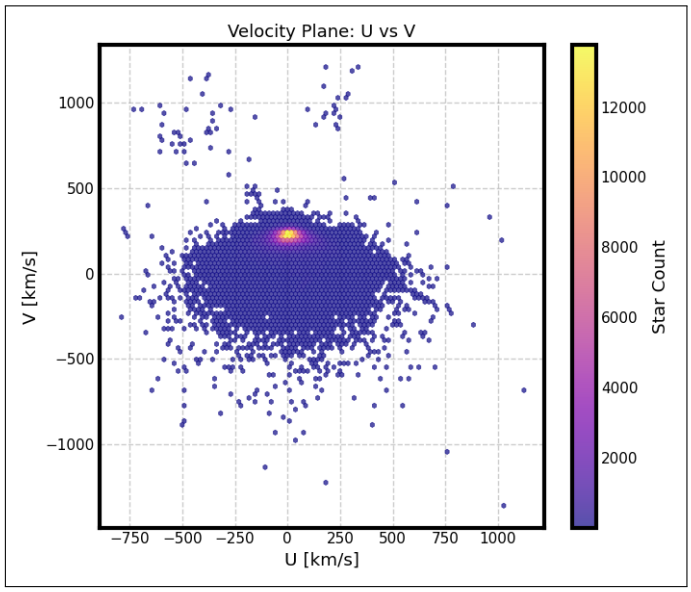


Fig. 11. UV cut of the velocity ellipsoid.

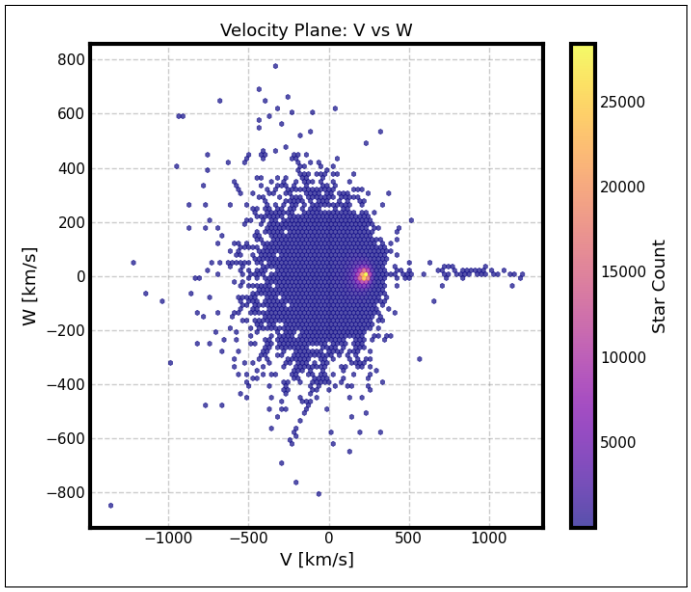


Fig. 12. VW cut of the velocity ellipsoid.

In the **UV plane**, the velocity ellipsoid exhibits a clearly elongated and tilted distribution. The central overdensity is centred around $U \approx 0$ km/s and $V \approx 220$–240 km/s, consistent with expectations for the Local Standard of Rest. The tilt of the ellipsoid away from the U axis reflects a non-zero covariance between the U and V components, indicative of asymmetric drift and radial streaming motions. The vertex deviation, computed from the

covariance matrix, yields a mean angle of approximately $\theta \approx$ -26.64°.This value represents the inclination of the major axis of the velocity ellipsoid relative to the U axis, and is consistent with previous studies that report vertex deviations ranging from $-15°$ to $-30°$ for local disk populations [28].

The **VW projection** displays a more symmetric, vertically elongated structure. The distribution is centred around $W \approx 0$ km/s, confirming that most stars oscillate near the Galactic plane. The lack of tilt in this plane suggests negligible vertical streaming relative to the Galactic rotation.

In the **UW plane**, the distribution shows a slightly diamond-like shape with a broader dispersion in U than in W. No significant inclination is visible, and the ellipsoid appears symmetric across both axes, supporting the assumption that U and W are dynamically uncorrelated for the majority of the sample.

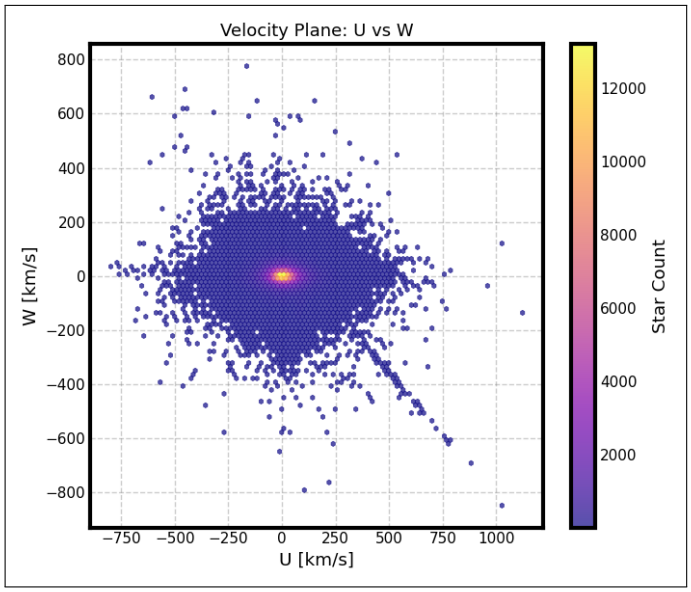


Fig. 13. UW cut of the velocity ellipsoid.

D. Velocity distribution analysis

To explore spatial variations in stellar kinematics and analyse the distribution of UV velocities, the sample was divided into square bins of size 1 kpc in the Galactocentric XY plane. This bin size was chosen to ensure a balance between spatial resolution and statistical significance. Only bins containing more than 20 stars were considered to ensure statistical robustness.

From this, the vector field (representing the mean radial, U, and azimuthal, V, velocity components) for stars within spatial bins across the XY plane and the standard deviation of the radial velocity component (σ_U) and the azimuthal velocity component (σ_V) were calculated independently.

D.1. Mean U and V Velocities Across the Galactic Plane

The resulting velocity field (**Figure 14**) displays a globally coherent rotational pattern, with stars in the outer disk exhibiting a systematic trend toward positive V and small U values, consistent with Galactic rotation. The direction and magnitude of the vectors reflect the underlying disk kinematics: near the solar radius, stars follow nearly circular orbits with dominant V motion and minimal radial excursions ($U \approx 0$ km/s). In the inner Galaxy ($X < 8$ kpc), the field shows a mild retrograde

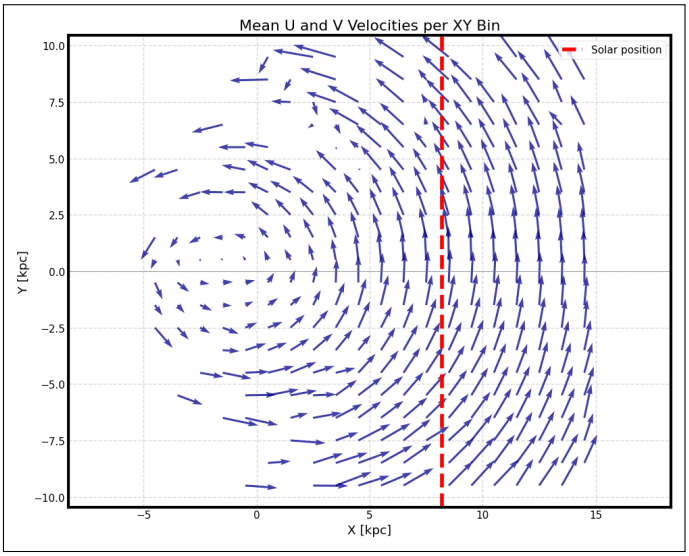


Fig. 14. Vector field for stars within spatial bins across the XY plane. The red dashed line indicates the Galactocentric X-position of the Sun ($X \approx 8.2$ kpc).

asymmetry in V, suggesting the influence of non-axisymmetric structures or asymmetric drift effects.

Radial streaming motions are also visible, because the average radial velocity component $\langle U \rangle$ is not close to zero, resulting in regions where the arrows point outward from the Galactic centre ($U > 0$) or inward ($U < 0$) reflect systematic flows away from or toward the centre, respectively.

D.2. Velocity Dispersion Maps in the XY Plane

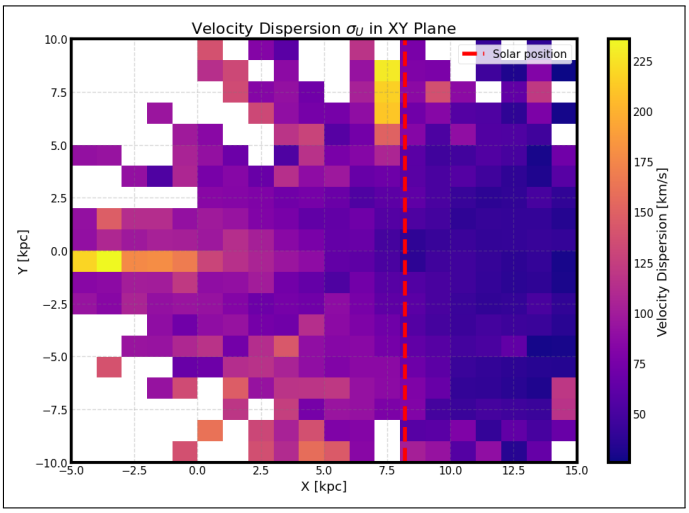


Fig. 15. Velocity Dispersion σ_U in XY Plane. The solar Galactocentric position ($X \approx 8.2$ kpc, $Y \approx 0$ kpc) is marked with a red dashed line for reference.

Figures 15 and 16 display the resulting maps of σ_U and σ_V, respectively.

The map of σ_U shows a clear gradient in the radial velocity dispersion, with values increasing toward the Galactic centre and remaining elevated in inner disk regions. The highest dispersions are found within $X \lesssim 0$–2 kpc, where σ_U exceeds 200

km/s in some bins. This is consistent with the presence of dynamically hotter populations such as thick disk or halo stars in these regions. In the outer disk, σ_U decreases smoothly, reaching values below 50 km/s beyond $X > 10$ kpc, indicating a colder kinematic regime.

The map of σ_V follows a similar trend, with elevated values near the inner Galaxy and lower dispersions in the outer disk. However, the distribution of σ_V is more spatially complex and asymmetric, with some localized enhancements beyond the solar radius. These may be associated with substructures, asymmetric drift, or field selection effects. A clear transition from high to low σ_V values is observed at approximately the solar circle, highlighting the change from a dynamically hotter central region to a more kinematically relaxed outer disk.

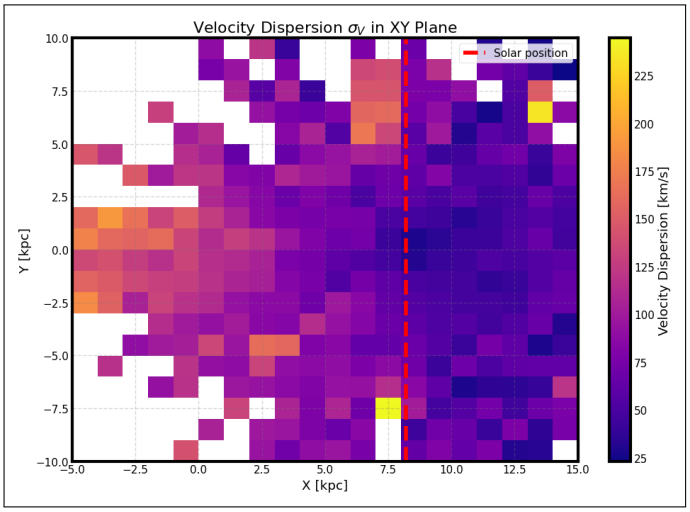


Fig. 16. Velocity Dispersion σ_V in XY Plane. The solar Galactocentric position ($X \approx 8.2$ kpc, $Y \approx 0$ kpc) is marked with a red dashed line for reference.

E. Kinematic Trends in the Solar Neighbourhood by Metallicity and Alpha Enhancement

To investigate the kinematic properties of different stellar populations in the local Galactic environment, all stars within 250 pc of the Sun were selected, resulting in a subsample of 28,157 objects. The radial (U) and azimuthal (V) components of the velocity were then plotted in the UV plane. The sample was subdivided into discrete bins of chemical abundance: three intervals in [Fe/H] (**Figures 17**, **18**, **19**) and three in [α/Fe] (**Figures 20**, **21**, **22**), using the APOGEE parameters `FE_H` and `ALPHA_M`, respectively. For each bin, a 2D histogram of U vs. V was constructed using the `hexbin` function from `matplotlib`, which allows for density-based visualization of stellar velocities.

The three metallicity bins reveal clear trends in rotational support and velocity dispersion:

- **[Fe/H] $\in$ [–1.5, –0.5]**: This metal-poor group displays a relatively broad distribution in both U and V, with most stars rotating at or above $V \sim 220$–250 km/s, but with substantial scatter. The wider spread in U suggests a dynamically hotter population, possibly associated with the thick disk or even halo contaminants.

- **[Fe/H] $\in$ [–0.5, 0.0]**: The bulk of stars lie in this bin, with a concentrated peak near $(U, V) = (0, 240)$ km/s and a more symmetric and narrower velocity ellipsoid. This group likely represents the chemically well-mixed thin disk, showing cold rotational kinematics and minimal asymmetric drift.

- **[Fe/H] $\in$ [0.0, 0.5]**: The metal-rich population also exhibits a cold, symmetric distribution, though the peak of V is slightly lower ($\sim$ 220–230 km/s), possibly indicating mild asymmetric drift. The velocity ellipsoid is more rounded compared to the intermediate bin.

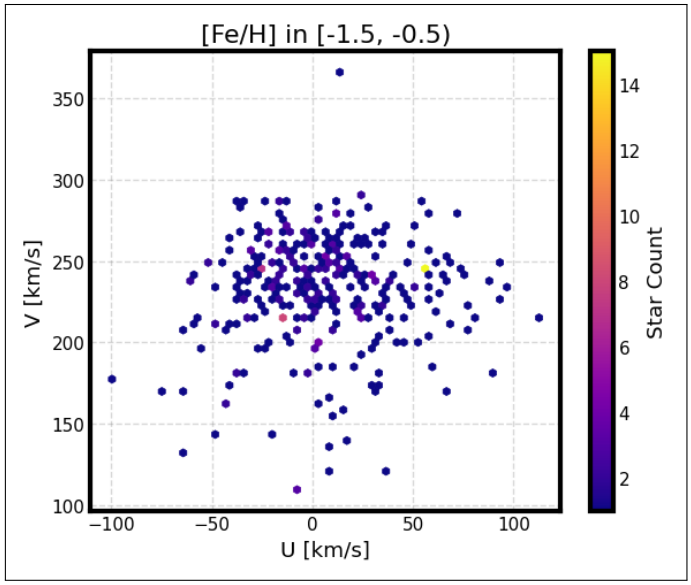


Fig. 17. Velocity distribution in the UV plane examined in **[Fe/H] $\in$ [–1.5, –0.5]**.

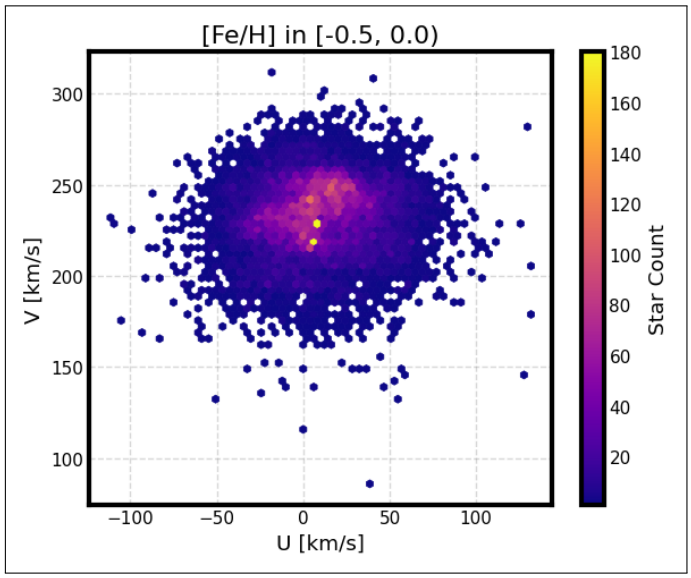


Fig. 18. Velocity distribution in the UV plane examined in **[Fe/H] $\in$ [–0.5, 0.0]**.

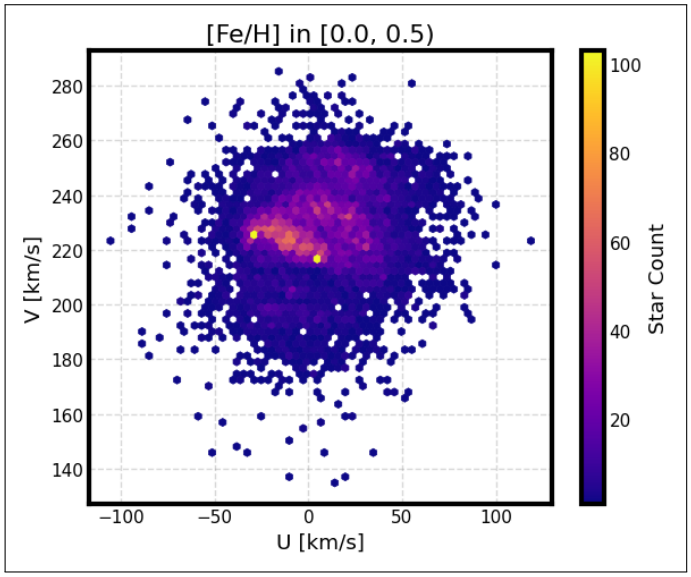


Fig. 19. Velocity distribution in the UV plane examined in **[Fe/H] ∈ [0.0, 0.5]**.

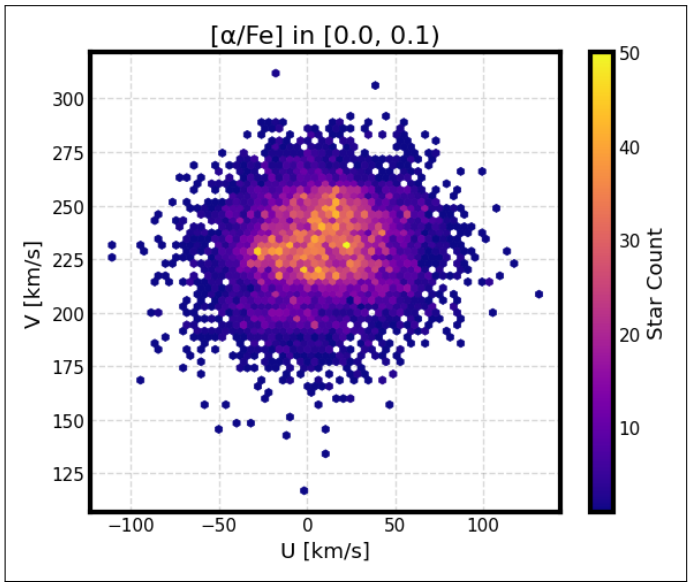


Fig. 20. Velocity distribution in the UV plane examined in **[α/Fe] ∈ [0.0, 0.1]**.

In terms of alpha-element abundance:

- **[α/Fe] ∈ [0.0, 0.1]**: This low-α population exhibits a tight, symmetric ellipsoid in the UV plane centered near $V \sim$ 230–250 km/s, characteristic of a thin disk dominated group with minimal velocity dispersion. These stars are typically younger and dynamically colder.

- **[α/Fe] ∈ [0.1, 0.25]**: Stars in this intermediate-α bin show a broader velocity distribution and a shift in the azimuthal velocity peak toward lower values ($V \sim$ 210–220 km/s). This is indicative of an older, kinematically warmer population associated with the thick disk. The increase in U dispersion further supports their dynamically heated nature.

- **[α/Fe] ∈ [0.25, 0.5]**: The most α-enhanced stars form a very sparse population in the solar vicinity, with only a few dozen stars. Despite the limited statistics, these stars span a wide range of U and V velocities, consistent with thick disk or even halo contamination. Their broader velocity ellipsoid and lower rotation support this interpretation, although caution is needed due to small number effects.

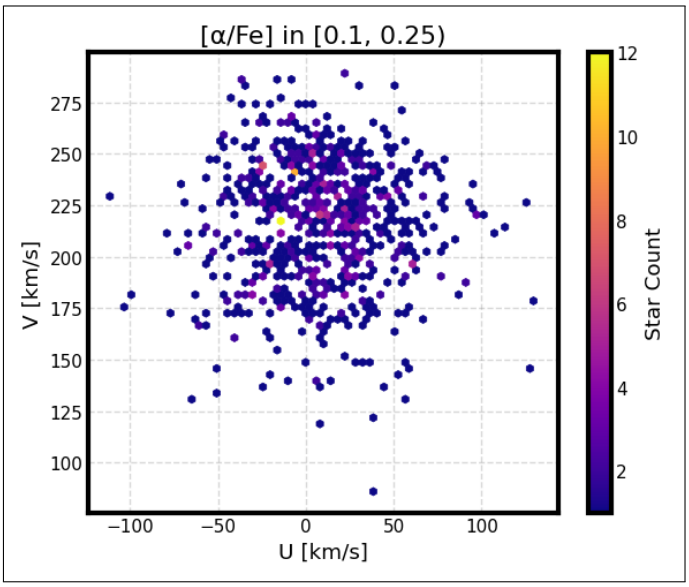


Fig. 21. Velocity distribution in the UV plane examined in **[α/Fe] ∈ [0.1, 0.25]**.

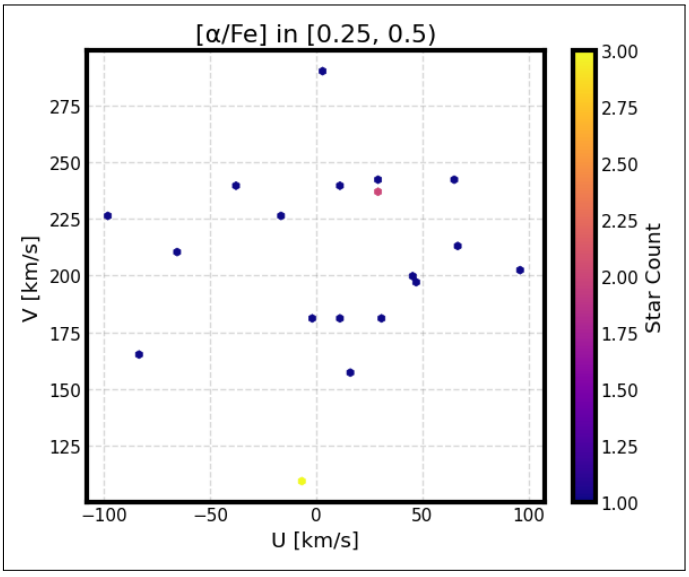


Fig. 22. Velocity distribution in the UV plane examined in **[α/Fe] ∈ [0.25, 0.5]**.

The overall trends align with expectations from Galactic chemical evolution: stars with lower [Fe/H] and higher [α/Fe] are typically older and dynamically hotter, exhibiting greater velocity dispersion and slower rotation. Conversely, metal-rich, alpha-poor stars are kinematically colder and more rotationally supported, consistent with a younger thin disk population.

F. Identification of Moving Groups in the Solar Neighbourhood

To identify kinematic groups in the solar vicinity, the velocity distribution in the UV plane was analysed for stars located within a radius of 250 pc from the Sun. A total of 28,157 stars were used for this clustering analysis. The `DBSCAN` algorithm (Density-Based Spatial Clustering of Applications with Noise) was employed to search for substructures within the local velocity space. DBSCAN is particularly well-suited for this task as it does not require a predefined number of clusters and can detect irregularly shaped groups.

The choice of parameters was guided by a k-distance graph constructed from the UV velocity space using the 10th nearest neighbour. The "knee" of the curve suggested an optimal value of $\epsilon \approx 3.5$ km/s. A minimum sample threshold of `min_samples` = 7 was selected to enhance the detection of less populated groups while reducing spurious detections (**Figure 23**).

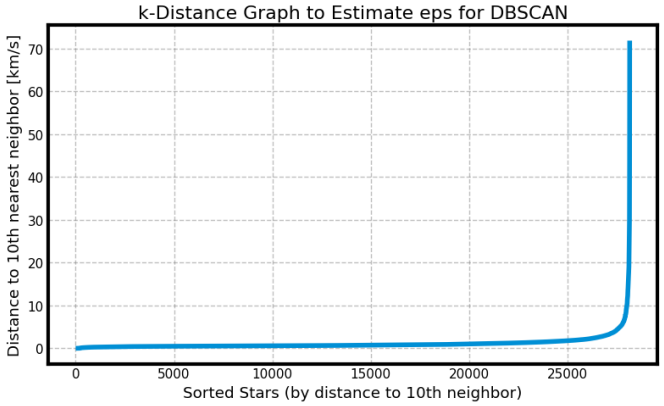


Fig. 23. K-distance curve (using the 10th nearest neighbour) used to estimate the optimal ϵ parameter for DBSCAN. The "elbow" near $\epsilon \approx 3.5$ km/s is adopted as the clustering threshold.

The resulting clustering map (Figure 24) displays the identified kinematic groups overlaid on the classical zones of known moving groups from the literature, such as Hyades, Hercules, Coma Berenices, Sirius, and Arcturus. The reference boundaries were adapted from [29], who delineated the approximate locations of these groups in the UV plane based on *Gaia* and RAVE data.

As seen in **Figure 24**, only one of the major groups—**Hercules**—was partially detected in this analysis. While other small clusters were formally identified, they do not overlap with the expected regions for classical moving groups. This suggests that the selected DBSCAN parameters, while statistically reasonable, may not capture the full extent of velocity substructure present in the data. Possible limitations include:

- The UV distribution is dominated by a dense core of stars near $(U, V) \sim (0, 220)$ km/s, which may blur group boundaries.
- The algorithm's sensitivity to noise and uneven density can lead to the merging or fragmentation of real groups.
- Projection effects and measurement uncertainties, particularly in proper motions, may affect cluster separability.

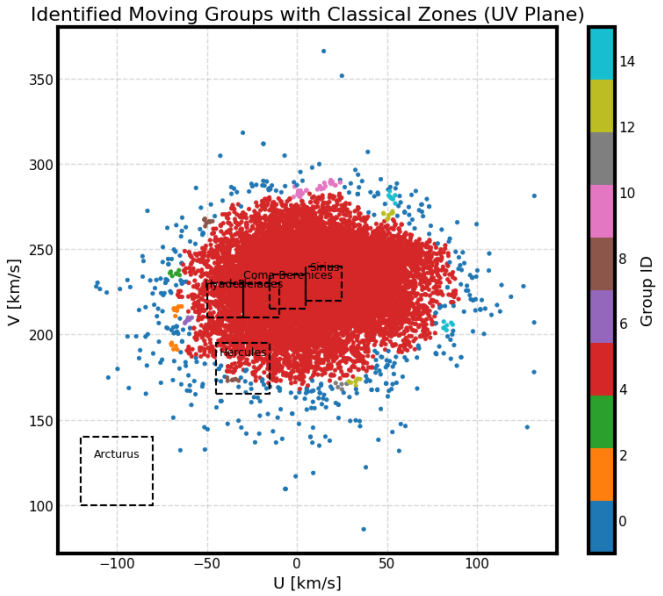


Fig. 24. Identified moving groups in the UV velocity plane using DBSCAN ($\epsilon = 3.5$ km/s, `min_samples` = 7), overlaid with classical moving group zones adapted from [29].

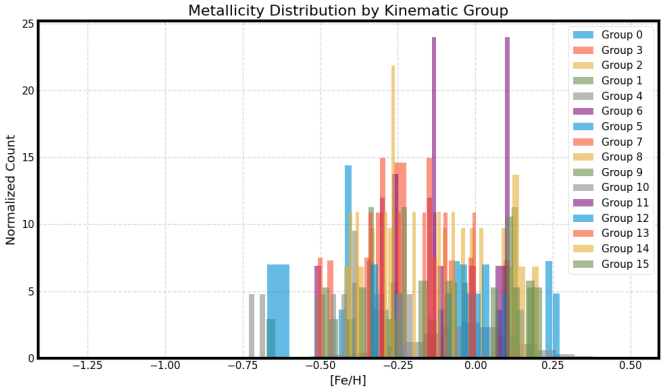


Fig. 25. Normalised [Fe/H] distribution for each kinematic group identified by DBSCAN. The metallicity range suggests a diversity of stellar populations and formation histories.

The metallicity distribution for each identified group is shown in Figure 25. The groups exhibit a broad range of [Fe/H] values, with peaks typically between −0.4 and 0.0 dex. This diversity is consistent with the idea that moving groups are not necessarily chemically homogeneous, and may arise from resonances, perturbations, or dissolution of open clusters. However, due to the limited number of groups confidently associated with known structures, caution is warranted in over-interpreting the chemical trends.

G. Galactic Rotation Curve

The galactic rotation curve presented in Figure 26 was constructed by computing the median azimuthal velocity component V_ϕ of stars within galactocentric cylindrical shells spanning a radial range of 6 to 12 kpc. The sample was restricted to stars located near the Galactic plane, specifically those with $|Z| < 0.5$ kpc, to minimise the impact of vertical kinematics. The azimuthal velocity V_ϕ was derived from the computed U,

V, and W heliocentric velocity components, corrected for the motion of the Sun relative to the Local Standard of Rest (LSR) using the values $U_\odot = 11.1\,\mathrm{km\,s^{-1}}$, $V_\odot = 12.24\,\mathrm{km\,s^{-1}}$, and $W_\odot = 7.25\,\mathrm{km\,s^{-1}}$ [30].

The resulting velocity V_ϕ was shifted to the Galactocentric frame by subtracting the circular velocity at the solar radius, assumed to be $V_{\mathrm{LSR}} = 220\,\mathrm{km\,s^{-1}}$ [31]. Subsequently, V_ϕ was binned as a function of Galactocentric radius R_{GC} using bins of width 0.2 kpc, and the median V_ϕ was computed in each bin to trace the rotation profile.

The resulting curve exhibits a general flatness beyond $\sim$ 8.5 kpc, consistent with canonical models of the Milky Way rotation curve. However, a sharp and unexpected rise is observed near $R_{\mathrm{GC}} \sim 6.5$–8.0 kpc. This feature may be attributed to several factors: (i) low-number statistics or increased contamination in the innermost bins; (ii) residual systematic errors in the coordinate transformations or improper LSR corrections; or (iii) unmodelled non-axisymmetric structures inducing kinematic perturbations at these radii.

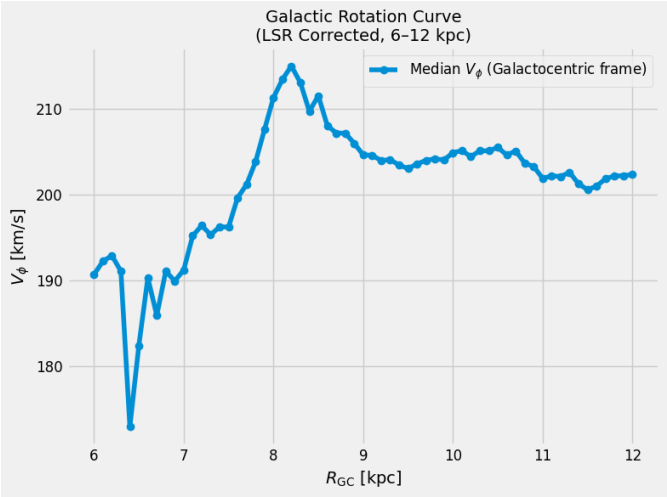


Fig. 26. Galactic rotation curve derived from stars between 6 and 12 kpc. The plot shows the median V_ϕ per radial bin.

H. Spatial distribution of oort constants

To investigate the local kinematics of the Milky Way, two-dimensional spatial maps of the Oort constants A, B, C, and K were constructed within a 2 kpc radius around the Sun in the Galactic plane. These constants were computed by fitting the linearised expressions of the velocity field to the observed stellar motions within square spatial bins. Each bin contained a sufficient number of stars to ensure statistical robustness, with a trade-off between spatial resolution and noise managed via the chosen `grid_size` parameter in the binning procedure.

The resulting maps show substantial noise and spatial fluctuations, with some bins displaying unphysically large or negative values for the Oort constants, particularly in low-density regions. This behaviour reflects both the inhomogeneous sampling of stars in the Gaia DR3–APOGEE-2 crossmatch and limitations in the local linear approximation, which breaks down when stellar velocities are sparsely or anisotropically distributed. Furthermore, outliers or stars with large uncertainties in distance or velocity components may disproportionately affect the regression results in certain bins.

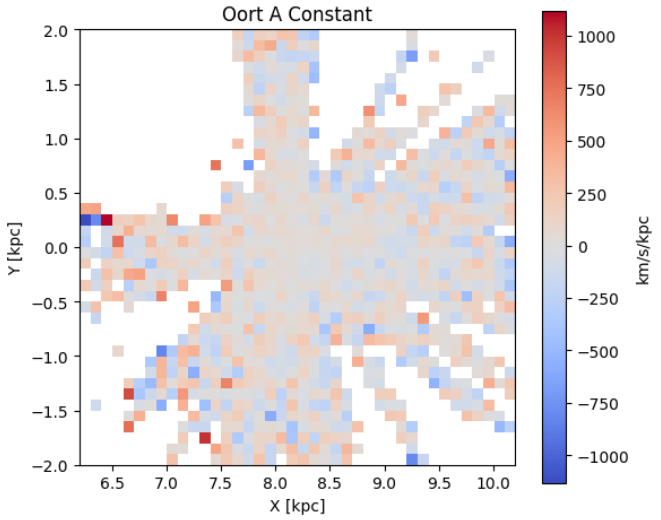


Fig. 27. Oort constant A map.

Notably, the central region surrounding the solar position tends to produce more coherent and physically plausible values, consistent with the expectations from prior literature: positive A and negative B in the solar vicinity imply differential rotation, while near-zero C and K suggest limited shear and divergence in the local velocity field. These findings are consistent with the velocity gradient patterns previously observed in the U–V vector fields and velocity dispersions analysed in earlier sections, lending qualitative support to the derived spatial structure.

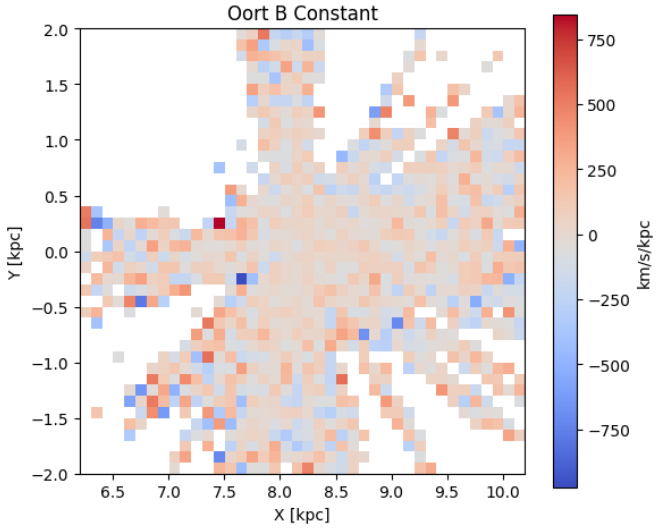


Fig. 28. Oort constant B map.

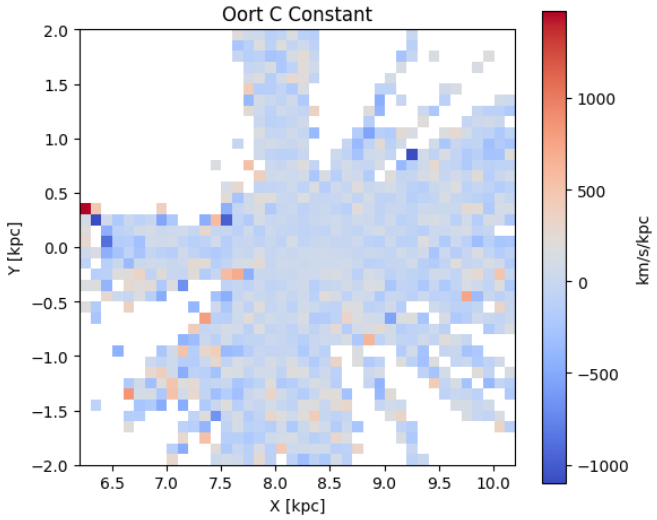


Fig. 29. Oort constant C map.

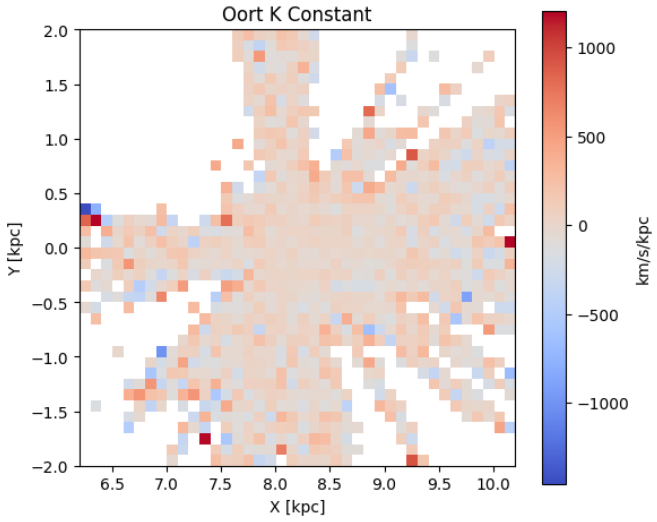


Fig. 30. Oort constant K map.

`Python` code is available here (bear in mind that the original file, with the filtered data, is needed to run it).

4. CONCLUSION

This project provides an integrated analysis of the structure and kinematics of the Milky Way based on high-quality Gaia DR3 and APOGEE-2 DR17 data. The first module establishes a structural classification of stellar populations using metallicity, spatial position, and vertical height, resulting in coherent subsets for the bulge, disks, halo, and solar neighbourhood. However, kinematic validation via the Toomre diagram reveals overlap between thin and thick disk populations, indicating that chemical and spatial criteria alone are insufficient for robust classification.

The second module incorporates full space velocities to explore the Galactic dynamical field in detail. Velocity ellipsoid projections reveal a significant vertex deviation and radial streaming motions, consistent with non-axisymmetric disk dynamics. The chemical dependence of the velocity distribution in the solar neighbourhood, the partial recovery of moving groups, and the shape of the rotation curve provide further insights into the structure and evolution of the Galaxy. While general trends conform to theoretical expectations, local perturbations and limitations in clustering performance point to the need for improved methodologies.

In summary, the results underscore the value of combining chemical, spatial, and kinematic diagnostics to disentangle the complex structure of the Milky Way. Future work should incorporate more sophisticated classification algorithms and modelling techniques to refine the identification of structural and dynamical subcomponents in the Galactic disk and halo.

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